DEMAE: Diffusion-Enhanced Masked Autoencoder for Hyperspectral Image Classification With Few Labeled Samples

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***Abstract*— Unlike other deep learning (DL) models, Transformer has the ability to extract long-range dependency features from hyperspectral image (HSI) data. Masked autoencoder (MAE), which is based on Transformer architecture, employs a "mask-reconstruction" strategy for training, allowing the model to be effective for downstream tasks. However, existing MAE-based methods only apply spectral or spatial masking to HSI and reconstruct them for feature learning, which is too simplistic and insufficient for the model to learn robust features. Additionally, the issue of lacking labeled samples in HSI and the primary objective of MAE to reduce the reliance on labeled samples are often overlooked. To address these issues, we are inspired by diffusion-based representation learning and propose diffusion-enhanced MAE (DEMAE) for HSI classification with few labeled samples. First, an asymmetric encoder–decoder framework is constructed as the backbone by stacking both conditional and standard Transformer blocks. Second, we devise an auxiliary task aimed at simultaneous denoising and reconstruction, facilitating heuristic feature learning from HSI data. Third, the encoder of DEMAE is isolated for training with few labeled samples. Finally, the encoder is used for classification, and a novel signal-to-noise ratio enhanced (SNR-Enhanced) loss function is introduced to regularize the model training process. The performance of DEMAE is evaluated on four benchmark datasets, demonstrating its superiority in classification accuracy and mapping capabilities on unlabeled areas compared to existing state-of-the-art methods with few labeled samples. The source code will be available online at https://github.com/ZhaohuiXue/DEMAE.**

***Index Terms*— Diffusion, few labeled samples, hyperspectral image (HSI) classification, masked autoencoder (MAE), Transformer.**

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I. Introduction

HYPERSPECTRAL image (HSI) captures reflectance information across hundreds of continuous spectral bands, enabling precise classification of land covers [1]. Therefore, HSI classification is often applied in various fields such as precision agriculture, mineral exploration, environmental monitoring [2], [3], [4], and [5].

Over the past decade, traditional machine learning (ML) methods and deep learning (DL) methods have significantly promoted the development of HSI classification. Among them, convolutional neural network (CNN) is the predominant DL framework, with research mainly focusing on 3DCNN [6], [7], [8]. However, CNN inherently suffers from the limitation of a finite receptive field. Given the complexities of HSI including mixed pixels, noise, ground object boundaries, conventional convolution operations struggle to extract typical features and suppress noise, resulting in the weakness of classification performance.

Compared with CNN, Transformer, which is based on attention mechanism [9], has the ability to enhance the weight of informative features while weakening the influence of noisy elements. Moreover, it is also better at modeling long-distance dependencies and exhibits remarkable versatility in handling various types of data. Relevant research in recent years includes that He et al. [10] initially employed CNN for spatial features extraction and Transformer for capturing spectral relationships for HSI classification. Subsequently, Qing et al. [11] integrated spectral attention and self-attention to improve Transformer's classification performance. Different from previous works, Hong et al. [12] proposed spectralformer (SF) as a pure Transformer architecture model for HSI classification, focusing on serializing spectral bands. However, these early methods all treated individual HSI pixels or local spectral bands as the processing units, limiting the potential for further improvement in classification accuracy. To address this issue, Sun et al. [13] introduced a tokenizer to convert HSI into semantic tokens and proposed spectral–spatial feature tokenization Transformer (SSFTT), which significantly improved the classification accuracy. Additionally, our previous work [14], local Transformer with spatial partition restore network (SPRLT), splits the HSI patch into overlapping sub-patches in the spatial dimension and performed self-attention within each sub-patch to enhance spatial feature

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extraction capabilities, which mitigated the limitations of Transformer-based methods in capturing local features. Correspondingly, Mei et al. [15] proposed group-aware hierarchical Transformer (GAHT) from the perspective of the local relationships between HSI bands, which enhanced spectral feature extraction capabilities.

However, the above methods primarily concentrated on designing specific modules or improving model architecture to fit labeled training samples better for satisfactory classification performance, which failed to fully exploit the strong generalization capabilities of Transformer when pre-trained with extensive unlabeled data.

To address the issue of insufficient generalization ability in Transformer and other DL frameworks, self-supervised learning (SSL) [16] has been developed recently. This method involves setting up auxiliary tasks and performs feature learning on extensive unlabeled data to obtain an initial feature extraction model. Subsequently, the initial model is fine-tuned for specific downstream tasks. SSL can be primarily categorized into contrastive learning-based, predictive learning-based, and generative learning-based methods [17]. Contrastive learning-based methods, such as decoupled-and-coupled networks (dc-Net) [18] and self-supervised contrastive learning (SSCL) [19], process an image through various transformations and train models by measuring the differences of image features extracted by different processing methods. Predictive learning-based methods focus on training models from the perspective of knowledge distillation, such as predicting the relative positions of image patches or the rotation angles between images [20]. Among generative learning-based methods, masked autoencoder (MAE) stands out as a widely researched generative SSL method built upon Transformer architecture, which is trained by randomly masking image patches and reconstructing them [21]. Recently, several studies utilizing MAE technique have emerged in the field of HSI classification. Ibaez et al. [22] earlier introduced masked autoencoding spectral–spatial Transformer (MAEST), which established a spectral–spatial reconstruction task for pre-training. Similarly, Liu et al. [17] proposed SSL of spectral masking (SSLSM) for training Transformer-based model. Different from the aforementioned "pre-training and fine-tuning" approach, Huang et al. [23] incorporated additional contrastive learning and reconstruction tasks during the fine-tuning stage to enhance feature extraction capabilities. Subsequently, considering the popularity of both contrastive learning and MAE within SSL, Cao et al. [24] merged them and proposed Transformer-based MAE using contrastive learning (TMAC). Additionally, Zhou et al. [25] proposed masked spectral–spatial feature prediction (MSSFP) aimed at enhancing the Transformer model's capability to extract features from complex spectral–spatial structures in HSI. Very recently, Hong et al. [26] adopted a 3-D masking strategy based on MAE technique to train SpectralGPT foundation model for a variety of downstream tasks, which allows the model to flexibly handle diverse inputs in terms of size, resolution, temporal variability, and geographical coverage.

Although existing MAE-based methods have achieved better classification performance compared to Transformer-based methods, their training approaches are similar, which merely apply the traditional reconstruction tasks for model training, limiting the ability to extract typical features of HSI. In addition, most methods still conduct experiments under the condition of a large number of labeled samples, ignoring the issue of insufficient labeled samples available for training due to the low efficiency of manual labeling for HSI data.

To address the issue of limited feature extraction capability caused by single auxiliary task in existing MAE-based methods for HSI classification, we draw inspiration from the training strategy of diffusion models [27], which is founded upon parameterized Markov chains and utilizes multistep prediction for image generation, and consider adopting a novel training approach from the perspective of representation learning to enhance the model's understanding of HSI. Recent studies utilizing diffusion models to learn image feature representation include the following. Preechakul et al. [28] earlier proposed diffusion autoencoders (DAEs) to extract deep semantic and decodable features, utilized for image classification and near-exact reconstruction respectively. Most of the subsequent research employed the diffusion model for downstream tasks from the perspective of SSL. Xiang et al. [29] proposed denoising DAEs (DDAEs), trained using a generative pre-training method, capable of extracting discriminative features for classification from the middle layer of the model. Different from conventional SSL methods employing reconstruction tasks, Hudson et al. [30] conducted SSL through image generation from varied perspectives, enabling to extract visual semantic features in an unsupervised manner. Hu et al. [31] pointed out that extracting guidance signals from the diffusion model is equivalent to extracting features from class-conditioned diffusion model, and employing regularization methods can enhance features discriminability. More recently, Chen et al. [32] deconstructed denoising diffusion model (DDM) and identified that DAE plays a crucial role in extracting high-quality features. The methods mentioned above all rely on UNet architecture, while Transformer-based studies are relatively rare, mainly comprising [33], [34], and [35]. Moreover, in the realm of image segmentation, relevant studies mainly encompass techniques based on diffusion feature extraction [36] and diffusion pre-training [37]. However, current diffusion-based studies for HSI processing are relatively rare, merely including HSI super-resolution [38], HSI reconstruction [39], HSI fusion [40], HSI clustering [41], HSI unmixing [42], HSI change detection [43], and HSI classification [44], [45]. Consequently, there remains a need for further research to enhance diffusion-based DL models for a more comprehensive understanding of HSI data.

In this article, we propose diffusion-enhanced MAE (DEMAE) for HSI classification with few labeled samples. First, an asymmetric encoder–decoder architecture is designed to simplify the decoder, compelling the encoder to extract more robust features from the HSI data. Second, we establish a challenging auxiliary task that involves masking and noising operations on HSI data and perform a pre-training task of denoising and reconstruction simultaneously. This approach heuristically allows the model to autonomously learn generalization features. Third, to leverage the versatility of

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Transformer in processing various data types, only a few of masked labeled samples with noise are utilized for fine-tuning the pre-trained model. This strategy ensures rapid convergence and mitigates overfitting issues. Finally, considering the quality of the labeled samples during the fine-tuning stage, we introduce the signal-to-noise ratio (SNR) of the samples into the training process to standardize the iterative updates and improve the classification performance of the model.

In summary, the main contributions of our work are as follows.

1) We propose a novel SSL framework, DEMAE, for HSI classification with few labeled samples. Different from previous approaches, we devise a highly challenging auxiliary task during the pre-training stage to encourage DEMAE to learn robust features. And DEMAE exhibits superior performance across four datasets.
2) Leveraging the Transformer's versatility in handling diverse data types, we utilize masked samples with noise, which can be regarded as a defective but unique data type that is totally different from the regular HSI patch, during both the pre-training and fine-tuning stages. This approach alleviates the overfitting challenge of DL models with few labeled samples.
3) We introduce a conditional Transformer, which considers the quality of training samples, as the encoder of DEMAE. Additionally, a signal-to-noise ratio-enhanced (SNR-Enhanced) loss function, which incorporates sample quality information as a priori into the loss, is proposed to standardize the iterative update process of the model for classification.
4) We undertake an initial exploration of pre-training with unlabeled samples in the source-domain data followed by fine-tuning with a few labeled samples in the target-domain data between similar HSI scenarios, which is a heuristic cross-domain experiment aimed at fundamentally alleviating the issue of scarce labeled samples and improving the generalization ability of model.

II. Proposed Methodology

The proposed DEMAE comprises four stages: HSI data preprocessing, self-supervised pre-training, fine-tuning, and classification. An overview of the model is depicted in Fig. 1.

A. HSI Data Preprocessing Stage

In the data preprocessing stage, PCA whitening is employed to map HSI into a latent space, reducing model parameters and enhancing training speed. From a data distribution perspective, PCA whitening yields uncorrelated feature components with a mean of 0 and a variance of 1, which is similar to the characteristics of a multivariate standard Gaussian distribution. These properties facilitate the derivation of relevant formulas in subsequent steps.

Let $X \in \mathbb{R}^{h\times w\times c}$ denote the HSI patch centered on the pixel targeted for classification after PCA whitening, where h, w, and c represent the size hyperparameters. Then the HSI patch X is serialized into an HSI sequence along the spatial dimension, and random mask processing is applied to X based on a predefined ratio, resulting in the division into x^{vis} and x^{mask}. This operation ensures the inclusion of the center pixel of X in x^{vis}. We denote $x^{\text{vis}} = [x_1^{\text{vis}}; x_2^{\text{vis}}; \cdots; x_{N1}^{\text{vis}}]$ and $x^{\text{mask}} = [x_1^{\text{mask}}; x_2^{\text{mask}}; \cdots; x_{N2}^{\text{mask}}]$, where $N1$ represents the number of visible pixels, $N2$ represents the number of masked pixels in the HSI sequence. DEMAE conducts denoising task on the x^{vis} and reconstruction task on the x^{mask}.

Then noise addition operation is performed on x^{vis}. The forward process of the diffusion model employs a Markov chain for modeling. Note that the initial state x_0^{vis} is x^{vis} without noise, and $x_t^{\text{vis}} = (\alpha_t)^{1/2}x_{t-1}^{\text{vis}} + (\beta_t)^{1/2}\epsilon_{t-1}$ denotes the data obtained by weighting the signal and noise at the t level with $\epsilon_{t-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, where t is a scalar that is randomly generated during the pre-training and fine-tuning stages, α_t and β_t represent the weighting parameters for the signal and noise components in x_t^{vis}, satisfying $\alpha_t + \beta_t = 1$. Specifically, a linear schedule is employed to calculate the noise content at each noise level t, defined as $\alpha_t = (T - t/T)\alpha_0 + \frac{t}{T}\alpha_T$, where $\alpha_0 = 0.9999$, $\alpha_T = 0.98$, and T is a hyperparameter representing the total number of noise levels. Therefore, the forward noise addition process is recorded as $q(x_t^{\text{vis}} \mid x_{t-1}^{\text{vis}}) = \mathcal{N}(x_t^{\text{vis}}; (\alpha_t)^{1/2}x_{t-1}^{\text{vis}}, (1-\alpha_t)^{1/2}\mathbf{I})$ $(t = 1, 2, \ldots, T)$, and the following formula can be derived based on the reparameterization trick and the principles of Markov chain and normal distribution:

$$\begin{aligned} x_{t+1}^{\text{vis}} &= \sqrt{\alpha_{t+1}}x_t^{\text{vis}} + \sqrt{1-\alpha_{t+1}}\epsilon_t \\ &= \sqrt{\alpha_t\alpha_{t+1}}x_{t-1}^{\text{vis}} + \sqrt{(1-\alpha_t)\alpha_{t+1}}\epsilon_{t-1} \\ &\quad + \sqrt{1-\alpha_{t+1}}\epsilon_t \end{aligned} \tag{1}$$

$$q\left(x_{t+1}^{\text{vis}} \mid x_{t-1}^{\text{vis}}\right) = \mathcal{N}\left(x_{t+1}^{\text{vis}}; \sqrt{\alpha_t\alpha_{t+1}}x_{t-1}^{\text{vis}}, (1-\alpha_t\alpha_{t+1})\mathbf{I}\right). \tag{2}$$

And, it can be summarized that $q(x_t^{\text{vis}} \mid x_0^{\text{vis}}) = \mathcal{N}(x_t^{\text{vis}}; (\bar{\alpha}_t)^{1/2}x_0^{\text{vis}}, (1-\bar{\alpha}_t)\mathbf{I})$ based on (2), where $\bar{\alpha}_t = \prod_{i=0}^{t}\alpha_i$.

B. Self-Supervised Pre-Training Stage

DEMAE treats each pixel as a token in the HSI sequence. Additionally, a class token x_{class} associated with the classification and a time token x_{time} associated with the diffusion process are introduced at the beginning of the sequence. Among them, the weight of x_{class} is random initialized, while the weight of x_{time} is initialized based on the noise level t according to the following formula:

$$\begin{aligned} x_{\text{time}} = \Bigg[&\sin\left(t \times \exp\left(i \times \frac{-\log(10000)}{d_{\text{time}}/2 - 1}\right)\right) \\ &\cos\left(t \times \exp\left(j \times \frac{-\log(10000)}{d_{\text{time}}/2 - 1}\right)\right)\Bigg] \end{aligned} \tag{3}$$

where d_{time} represents the feature dimension of the initial x_{time}, and i ranges from 0 to $d_{\text{time}}/2 - 1$, j ranges from $d_{\text{time}}/2$ to $d_{\text{time}} - 1$. Subsequently, x_{time} undergoes a two-layer MLP embedding and is then incorporated into the sequence.

As we entirely abandon convolution operations, we utilize position embedding to inform the model about the absolute position of each pixel in the HSI sequence, which can be

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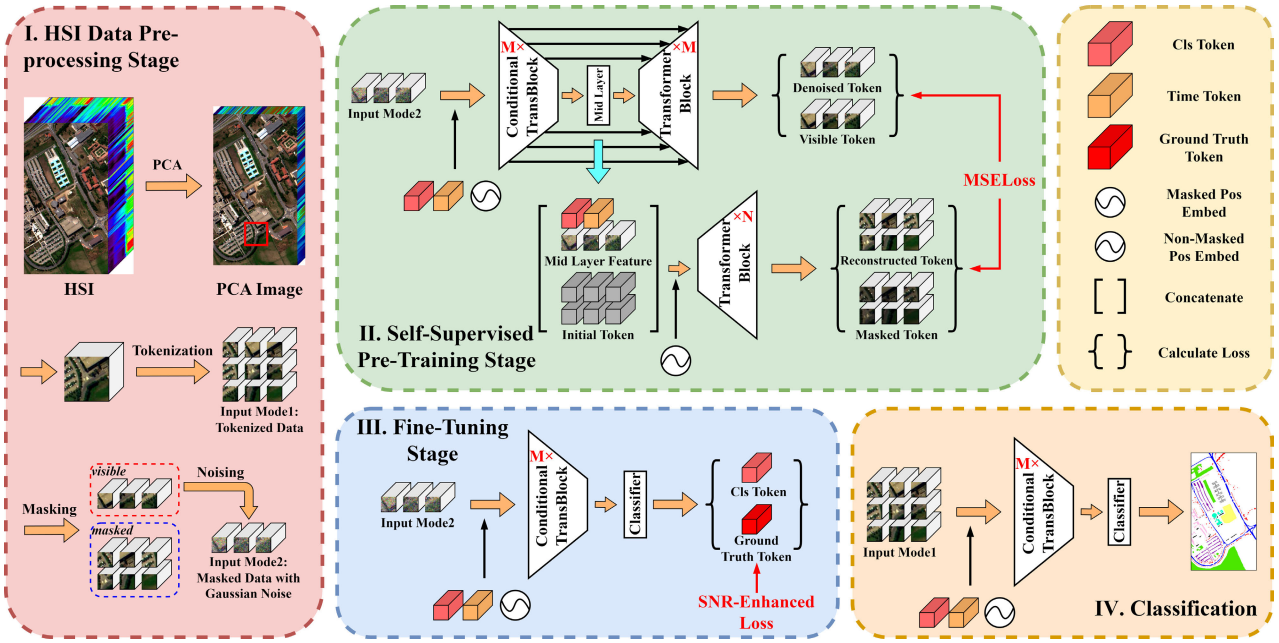


Fig. 1. Graphical illustration of the proposed DEMAE.

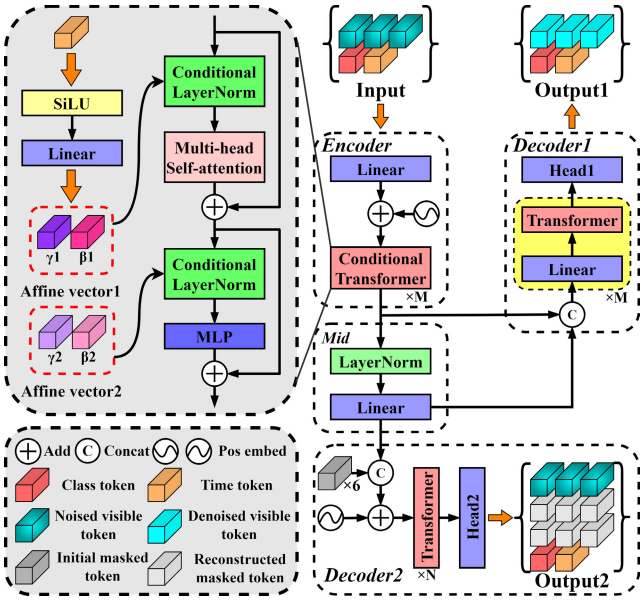


Fig. 2. Pre-training details of the proposed DEMAE.

represented by the following formula:

$$E_{\mathrm{pos}(p,2i)} = \sin\left(\frac{p}{10000^{2k/d_{\mathrm{model}}}}\right) \tag{4}$$

$$E_{\mathrm{pos}(p,2i+1)} = \cos\left(\frac{p}{10000^{2k/d_{\mathrm{model}}}}\right) \tag{5}$$

where p denotes the position of the current pixel in the HSI sequence, k represents the component of the kth dimension in the position embedding and ranges from 0 to $d_{\mathrm{model}}/2-1$.

1) Conditional Transformer-Based Encoder: Conditional modules are more commonly employed in the realm of image generation. Generally, an additional label is encoded as a guiding condition and embedded within the model to regulate the generated images. In DEMAE, the relationship between x_{time} and two learnable affine transform parameters, scale and bias, in layer normalization is established, instructing the model to employ distinct normalization methods to transform feature for HSI sequences with varying noise contents. The specifics of this module in the encoder are illustrated in Fig. 2. Moreover, it can also be represented by the following formula:

$$z_{EN(0)} = \left[x_{\mathrm{class}}; x_{\mathrm{time}}; x^{\mathrm{vis}}_{1-t}E; x^{\mathrm{vis}}_{2-t}E; \cdots; x^{\mathrm{vis}}_{N1-t}E\right] + E^{\mathrm{vis}}_{\mathrm{pos}} \tag{6}$$

$$z'_{EN(m)} = \mathrm{Attention}\left(\mathrm{cNorm}\left(z_{EN(m-1)}, x_{\mathrm{time}}\right)\right) + z_{EN(m-1)} \tag{7}$$

$$z_{EN(m)} = \mathrm{FFN}\left(\mathrm{cNorm}\left(z'_{EN(m)}, x_{\mathrm{time}}\right)\right) + z'_{EN(m)} \tag{8}$$

where x^{vis}_{p-t} denotes the pth HSI token in x^{vis}, E denotes the initial linear embedding layer of DEMAE, and $E^{\mathrm{vis}}_{\mathrm{pos}}$ represents the position embedding of the visible tokens. The variable m ranges from 1 to M, where M is the number of layers in the encoder, and cNorm$(\cdot)$ in (7) refers to the conditional layer normalization within the conditional Transformer. Furthermore, the middle layer connecting the encoder and decoder can be represented by the following formula:

$$z_{\mathrm{MID}} = \mathrm{Linear}\left(\mathrm{Norm}\left(z_{EN(M)}\right)\right) \tag{9}$$

where Linear$(\cdot)$ represents a linear layer with same-dimension embedding.

2) Diffusion Decoder and Denoising Loss Function: The conditional Transformer-based encoder and diffusion decoder are utilized to conduct the reverse denoising process. The diffusion decoder can be represented by the following formula:

$$z_{DE1(0)} = z_{\mathrm{MID}} \tag{10}$$

$$z'_{DE1(m)} = \mathrm{Linear}\left(\left[z_{DE1(m-1)}, z_{EN(M-m+1)}\right]\right) \tag{11}$$

$$z''_{DE1(m)} = \mathrm{Attention}\left(\mathrm{Norm}\left(z'_{DE1(m)}\right)\right) + z'_{DE1(m)} \tag{12}$$

$$z_{DE1(m)} = \mathrm{FFN}\left(\mathrm{Norm}\left(z''_{DE1(m)}\right)\right) + z''_{DE1(m)} \tag{13}$$

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where the linear layer in (11) embeds the feature dimension to half of the original input, and the value of m remains consistent with the encoder.

The reverse process, denoted by $p_\theta(x_{t-1}^{\text{vis}} \mid x_t^{\text{vis}})$, is also modeled by a Markov chain, where θ represents the learnable parameter of the denoising model. For the x_{t-1}^{vis} obtained through denoising x_t^{vis}, we ensure that this process can eliminate the random noise, obeying a Gaussian distribution, added by the forward process. Therefore, for the objective optimization function of this branch, our aim is to seek the distributions of $p_\theta(x_{t-1}^{\text{vis}} \mid x_t^{\text{vis}})$ and $q(x_{t-1}^{\text{vis}} \mid x_t^{\text{vis}})$ to be as close as possible. To deduce $q(x_{t-1}^{\text{vis}} \mid x_t^{\text{vis}})$, we set the variance of p_θ equal with that of q at the same t, which is same with DDPM [27], and rely on Bayes' theorem, resulting in the following formula:

$$\begin{aligned} q\left(x_{t-1}^{\text{vis}} \mid x_t^{\text{vis}}, x_0^{\text{vis}}\right) &= \frac{q\left(x_t^{\text{vis}} \mid x_{t-1}^{\text{vis}}, x_0^{\text{vis}}\right) q\left(x_{t-1}^{\text{vis}} \mid x_0^{\text{vis}}\right)}{q\left(x_t^{\text{vis}} \mid x_0^{\text{vis}}\right)} \\ &\propto \exp\left\{-\frac{1}{2}\left[\frac{-2\sqrt{\alpha_t}x_t^{\text{vis}}x_{t-1}^{\text{vis}} + \alpha_t\left(x_{t-1}^{\text{vis}}\right)^2}{1-\alpha_t} + \frac{\left(x_{t-1}^{\text{vis}}\right)^2 - 2\sqrt{\bar{\alpha}_{t-1}}x_0^{\text{vis}}x_{t-1}^{\text{vis}}}{1-\bar{\alpha}_{t-1}}\right]\right\} \\ &\propto \mathcal{N}\left(x_{t-1}^{\text{vis}}; \frac{\sqrt{\alpha_t}\left(1-\bar{\alpha}_{t-1}\right)x_t^{\text{vis}} + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)x_0^{\text{vis}}}{1-\bar{\alpha}_t}, \frac{(1-\alpha_t)\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}\mathbf{I}\right). \end{aligned} \tag{14}$$

We introduce the scalar quantity $\sigma_q^2(t) = ((1-\alpha_t)(1-\bar{\alpha}_{t-1})/1-\bar{\alpha}_t)$ and the corresponding covariance matrix $\Sigma_q(t) = \sigma_q^2(t)\mathbf{I}$. The posterior probability distribution of the forward process is defined as $q(x_{t-1}^{\text{vis}} \mid x_t^{\text{vis}}, x_0^{\text{vis}}) = \mathcal{N}(x_{t-1}^{\text{vis}}; \mu_q, \Sigma_q(t))$, where μ_q is a function of x_t^{vis}, x_0^{vis} and t. And the probability distribution of the reverse process is established as $p_\theta(x_{t-1}^{\text{vis}} \mid x_t^{\text{vis}}) = \mathcal{N}(x_{t-1}^{\text{vis}}; \mu_p, \Sigma_q(t))$, where μ_p is a function of x_t^{vis} and t. This establishment is based on the assumption that the variance of the reverse process is equal to that of the forward process when the states are identical. We write μ_q as a function of x_t^{vis} and t, and similarly define μ_p in the form of μ_q based on (14), such that $\mu_p(x_t^{\text{vis}}, t) = (1/(\alpha_t)^{1/2})[x_t^{\text{vis}} - (1-\alpha_t/(1-\bar{\alpha}_t)^{1/2})\hat{\epsilon}_\theta(x_t^{\text{vis}}, t)]$. Subsequently, we utilize the KL divergence to formulate the optimization objective function as

$$\begin{aligned} &\arg\min_\theta D_{KL}\left(q\left(x_{t-1}^{\text{vis}} \mid x_t^{\text{vis}}, x_0^{\text{vis}}\right) \| p_\theta\left(x_{t-1}^{\text{vis}} \mid x_t^{\text{vis}}\right)\right) \\ &= \arg\min_\theta \frac{1}{2\sigma_q^2(t)}\left\|\mu_p - \mu_q\right\|_2^2 \\ &= \arg\min_\theta \frac{1}{2\sigma_q^2(t)} \cdot \frac{(1-\alpha_t)^2}{\alpha_t\left(1-\bar{\alpha}_t\right)}\left\|\epsilon - \hat{\epsilon}_\theta\left(x_t^{\text{vis}}, t\right)\right\|_2^2 \end{aligned} \tag{15}$$

where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The general diffusion model adopts $L_{\text{simple}}(\theta) = \|\epsilon - \hat{\epsilon}_\theta(x_t^{\text{vis}}, t)\|_2^2$ as the target loss function to train the model parameter θ for image generation. However, in DEMAE, it is necessary to ensure that the features model produced by the model can be used for HSI classification and correspond to the loss component of the MAE branch. Therefore, we aim for the loss to be a function about x_0^{vis}, which can be expressed as

$$\begin{aligned} L_{\text{simple}}(\theta) &= \left\|\epsilon - \hat{\epsilon}_\theta\left(x_t^{\text{vis}}, t\right)\right\|_2^2 \\ &= \left\|\frac{1}{\sqrt{1-\bar{\alpha}_t}}\left(x_t^{\text{vis}} - \sqrt{\bar{\alpha}_t}x_0^{\text{vis}}\right) - \frac{1}{\sqrt{1-\bar{\alpha}_t}}\left(x_t^{\text{vis}} - \sqrt{\bar{\alpha}_t}\hat{x}_\theta^{\text{vis}}\left(x_t^{\text{vis}}, t\right)\right)\right\|_2^2 \\ &= \frac{\bar{\alpha}_t}{1-\bar{\alpha}_t}\left\|\hat{x}_\theta^{\text{vis}}\left(x_t^{\text{vis}}, t\right) - x_0^{\text{vis}}\right\|_2^2 \end{aligned} \tag{16}$$

where $\hat{x}_\theta^{\text{vis}}(x_t^{\text{vis}}, t)$ represents the output of $z_{DE1(M)}$ after normalization and a linear layer. The denoising loss function of the diffusion branch is derived as

$$L_{\text{Diffusion}} = \left\|\hat{x}_\theta^{\text{vis}}\left(x_t^{\text{vis}}, t\right) - x_0^{\text{vis}}\right\|_2^2. \tag{17}$$

Given that $x_t^{\text{vis}} = (\bar{\alpha}_t)^{1/2}x_0^{\text{vis}} + (1-\bar{\alpha}_t)^{1/2}\epsilon$, $(\bar{\alpha}_t/1-\bar{\alpha}_t)$ represents the square of the ratio between the HSI sequence component and the noise component in x_t^{vis}. In DEMAE, we utilize the SNR defined as $\text{SNR}(t) = (\bar{\alpha}_t/1-\bar{\alpha}_t)$.

3) MAE Decoder and Reconstruction Loss Function: The conditional Transformer-based encoder is employed for feature learning on the visible HSI pixels, while the MAE decoder is tasked with reconstructing masked pixels. The feature output by the middle layer requires the attachment of $N2$ learnable initial tokens at the end of the HSI sequence. Subsequently, position embedding is applied to denote the absolute positions of all tokens. Then the full set is fed into the decoder for reconstruction task, which is stacked by Transformer block. The above process can be represented by the following formula:

$$z_{DE2(0)} = \left[zc; x_{1-\text{init}}^{\text{mask}}; x_{2-\text{init}}^{\text{mask}}; \cdots; x_{N2-\text{init}}^{\text{mask}}\right] + E_{\text{pos}}^{\text{full}} \tag{18}$$

$$z'_{DE2(n)} = \text{Attention}\left(\text{Norm}\left(z_{DE2(n-1)}\right)\right) + z_{DE2(n-1)} \tag{19}$$

$$z_{DE2(n)} = \text{FFN}\left(\text{Norm}\left(z'_{DE2(n)}\right)\right) + z'_{DE2(n)} \tag{20}$$

where $x_{p-\text{init}}^{\text{mask}}$ denotes the learnable initial token at position p in x^{mask}, $E_{\text{pos}}^{\text{full}}$ represents the position embedding sorted in order of visible and masked token positions, $n = 1, 2, \ldots, N$, and N is the number of layers in the MAE decoder. It is worth noting that the depth of MAE decoder does not necessarily match the depth of the encoder. This architecture ensures that the encoder can focus on developing robust feature extraction capabilities, rather than relying on the decoder to compensate for decoding capabilities.

Corresponding to the diffusion branch, the MAE branch solely calculates the reconstruction loss for the masked part in the HSI sequence. Note that $z_{DE2(N)}^{\text{mask}}$ represents the value of the masked position in $z_{DE2(n)}$, and $\hat{x}_\theta^{\text{mask}}$ is the output obtained after normalization and linear embedding operations on $z_{DE2(N)}^{\text{mask}}$. Therefore, the reconstruction loss of the MAE branch can be expressed by the following formula:

$$\hat{x}_\theta^{\text{mask}} = \text{Linear}\left(\text{Norm}\left(z_{DE2(N)}^{\text{mask}}\right)\right) \tag{21}$$

$$L_{\text{MAE}} = \left\|\hat{x}_\theta^{\text{mask}} - x^{\text{mask}}\right\|_2^2. \tag{22}$$

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4) Loss Function During the Pre-Training Stage: By combining the denoising and reconstruction losses obtained from the diffusion branch and the MAE branch, the loss function of DEMAE during the pre-training stage is derived as follows:

$$\begin{aligned} L_{\text{pre-train}} &= L_{\text{Diffusion}} + L_{\text{MAE}} \\ &= \left\| \hat{x}_\theta^{\text{vis}}\left(x_t^{\text{vis}}, t\right) - x_0^{\text{vis}} \right\|_2^2 + \left\| \hat{x}_\theta^{\text{mask}} - x^{\text{mask}} \right\|_2^2 \end{aligned} \tag{23}$$

C. Fine-Tuning Stage

During the fine-tuning stage, we extract the encoder and middle layer of DEMAE and connect them with a linear layer for classification. The data processing method during the fine-tuning stage mirrors that of the pre-training stage, including PCA whitening, serialization, random masking with center pixel preservation, and noising. However, the degree of masking and noising during the fine-tuning stage does not exceed that of the pre-training stage. Considering that DEMAE is designed to classify the center pixel of an HSI patch, we believe that masking and noising operations can enrich the diversity of feature representations for samples. The pre-trained model possesses feature extraction capabilities for such HSI samples, thereby increasing the possibility of reaching a better convergence point during the fine-tuning stage.

As for the loss function during the fine-tuning stage, we extract the class token output by the last linear layer and activate it through the softmax function to calculate the loss. Additionally, we introduce the SNR of the sample into the loss function, ensuring that the model assigns higher importance to samples with a high SNR, while imposing smaller penalties on samples with a lower SNR. The proposed SNR-Enhanced loss function can be expressed as follows:

$$L_{\text{SNREnhanced}} = -(1 - p_t)^{2\times \text{Sigmoid}(-\text{SNR}(t))} \log(p_t) \tag{24}$$

where p_t represents the probability that the sample is correctly classified.

D. Classification

During the testing phase, we only perform PCA whitening and serialization operations on the HSI patch and directly input it into the model backbone. Finally, the HSI patch is classified based on the output class token.

III. Experimental Results

A. Hyperspectral Datasets

We conduct experiments on four commonly used HSI datasets, including the University of Pavia (PU), Salinas (SA), University of Houston (HU), and WHU-LongKou (LK). Among them, bands absorbed by water vapor are specifically eliminated in the SA dataset. Corresponding size, band information, and number of labeled samples per class are reported in Table I.

B. Experimental Settings

1) Hyperparameters: The training epochs for each dataset during the pre-training and fine-tuning stages are set to (200, 180) for PU, (300, 120) for SA, (300, 160) for HU, and (200, 160) for LK. The batch size for each dataset is set to 512 during the pre-training stage, and this parameter is adjusted to 45 for PU and LK, 80 for SA, and 75 for HU during the fine-tuning stage. The learning rate for different datasets is set to 1e-3 during the pre-training stage, while the mask ratio is set to [0.25, 0.5, 0.75]. The range of noise scales is set to [1e-4, 2e-2] during both the pre-training and fine-tuning stages, which is consistent with DDPM [27]. In subsequent experiments, we will optimize other parameters, including the data dimension after PCA whitening operation, patch size, the number of noise levels $T_{\text{fine-tune}}$ during the pre-training stage and the number of noise levels $T_{\text{fine-tune}}$, mask ratio, and learning rate during the fine-tuning stage. Besides, the learning rate decay trick is applied during both the pre-training and fine-tuning stages, with a decay rate of 0.99 implemented at every epoch.

2) Training Samples: Unlabeled samples are utilized to train the initial weights of the model during the pre-training stage, aligning with the prior diffusion-based [44] and MAE-based [17], [24], [25] HSI classification methods. During the fine-tuning stage, [5, 10, 15, 20, 25] labeled samples from each class are randomly selected to compose the training set for HSI classification, and the remaining labeled samples constitute the test set for evaluating classification accuracy.

3) Comparison Methods: We select several representative HSI classification methods for comparison to assess the performance of the proposed DEMAE, including one classic CNN-based method (SSRN) [6], three few-shot learning-based methods (DPN-HRA) [46], (RN-FSC) [47], (S3Net) [48], one transfer learning-based method (DCFSL) [49], two Transformer-based methods (SSFTT) [13], (GAHT) [15], one MAE-based method (MAEST) [22], and one diffusion-based method (SpectralDiff) [44]. The parameter settings of the aforementioned comparative methods remain consistent with those reported in the original paper.

4) Running Platforms and Metrics: Except for DPN-HRA, which was run under the TensorFlow 1.15.5 and cuDNN 10.0 framework, all other methods were run under the PyTorch 1.10.2 and cuDNN 11.1.1 framework. Due to graphics memory constraints, experiments involving DCFSL and SpectralDiff were conducted using GeForce RTX 3090 (24 GB), while experiments for other methods were carried out using GeForce RTX 3080 (12 GB). Three commonly used indicators, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ), were utilized to assess the classification performance of different models. To ensure the reliability of the experimental results, we conducted ten independent experiments and computed the mean and standard deviation as the accuracy results.

C. Parameter Sensitivity Analysis

1) HSI Dimensionality Reduction: PCA whitening is used to reduce the dimensionality of HSI into [28, 32, 36, 40,

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TABLE I
SIZE, BAND INFORMATION, AND NUMBER OF LABELED SAMPLES PER CLASS FOR FOUR DATASETS

Datasets	PU		SA		HU		LK	
Size and bands	610×340×103		512×217×204		349×1905×144		550×400×270	
Class	Name	Number	Name	Number	Name	Number	Name	Number
1	Asphalt	6631	Brocoli green weeds 1	2009	Healthy grass	1251	Corn	34511
2	Meadows	18649	Brocoli green weeds 2	3726	Stressed grass	1254	Cotton	8374
3	Gravel	2099	Fallow	1976	Synthetic grass	697	Sesame	3031
4	Trees	3064	Fallow rough plow	1394	Trees	1244	Broad-leaf soybean	63212
5	Painted metal sheets	1345	Fallow smooth	2678	Soil	1242	Narrow-leaf soybean	4151
6	Bare soil	5029	Stubble	3959	Water	325	Rice	11854
7	Bitumen	1330	Celery	3579	Residential	1268	Water	67056
8	Self-Blocking Bricks	3682	Grapes untrained	11271	Commercial	1244	Roads and houses	7124
9	Shadows	947	Soil vinyard develop	6203	Road	1252	Mixed weed	5229
10			Corn senesced green weeds	3278	Highway	1227		
11			Lettuce romaine 4wk	1068	Railway	1235		
12			Lettuce romaine 5wk	1927	Parking Lot1	1233		
13			Lettuce romaine 6wk	916	Parking Lot2	469		
14			Lettuce romaine 7wk	1070	Tennis court	428		
15			Vinyard untrained	7268	Running track	660		
16			Vinyard vertical trellis	1807				
	Total	42776	Total	54129	Total	15029	Total	204542

44], followed by pre-training and fine-tuning stages. Fig. 3(a) illustrates the OA obtained by DEMAE under different parameters. Notably, setting this parameter to 36 for PU, HU, LK datasets, and to 40 for SA dataset, achieves higher classification accuracy.

2) Patch Size: We set this parameter to [7, 9, 11, 13, 15], and the corresponding OA obtained under different settings is shown in Fig. 3(b). When patch size is set to 11 × 11, the highest accuracy is achieved across all datasets. Therefore, we adopt this parameter setting for subsequent experiments.

3) Noise Levels $T_{pre-Train}$ During the Self-Supervised Pre-Training Stage: As the proposed DEMAE is primarily utilized for classification rather than generation tasks, it is unnecessary to ensure that data in high noise level states strictly obeys a standard normal distribution to facilitate generative tasks based on known noise distribution. Therefore, during the self-supervised pre-training stage, we set the number of noise levels $T_{\text{pre}-\text{train}}$ to a moderate range [100, 150, 200, 250, 300], ensuring that the noisy HSI data retains its original features rather than being overwhelmed by pure noise. Then, we temporarily set $T_{\text{fine}-\text{tune}}$ to half of $T_{\text{pre}-\text{train}}$, and the corresponding OA under different parameter settings is illustrated in Fig. 3(c). The classification accuracy is higher when $T_{\text{pre}-\text{train}}$ is set to 200 for PU and HU datasets representing urban scenes, and 300 for SA and LK datasets representing farmland scenes. In this context, we believe that the spectral differences between land covers in urban scenes are obvious, so lower noise levels is enough for the model to learn the distinctive features. On the contrary, the spectral differences between land covers in farmland scenes are subtle, and higher noise levels obscure the distinction between features, forcing the model to strengthen its ability to mine discriminative features from different land covers.

4) Mask Ratio During the Pre-Training Stage: We mask the HSI patch with ratios of [0.25, 0.5, 0.75] and calculate the reconstruction loss to train the proposed DEMAE. The OA obtained by the model under different parameter settings is shown in Fig. 3(d). Obviously, the highest classification accuracy is achieved across all datasets when the parameter is set to 0.75. Additionally, we found that a smaller mask ratio requires more training time during the pre-training stage because the model needs to process more HSI pixels, which slows down the forward propagation.

5) Noise Levels $T_{fine\text{-}Tune}$ During the Fine-Tuning Stage: In this part of the experiment, we set $T_{\text{fine}-\text{tune}}$ to start from 0 and increase at intervals of 50, while ensuring that $T_{\text{fine}-\text{tune}}$ does not exceed $T_{\text{pre}-\text{train}}$. As is shown in Fig. 3(e), the classification results obtained on PU and HU datasets exhibit greater sensitivity to this parameter than those on the other two datasets. Based on the above experimental results, we set $T_{\text{fine}-\text{tune}}$ to 150 for PU and SA, 50 for HU, and 100 for LK in the subsequent experiments.

6) Mask Ratio During the Fine-Tuning Stage: We decrease the mask ratio during the fine-tuning stage, setting this parameter to [0, 0.25, 0.5, 0.75] for the experiment. The corresponding results are shown in Fig. 3(f). In general, all datasets exhibit an upward trend in classification accuracy as the mask ratio increases. The trend is particularly obvious in PU and HU datasets representing urban scenes. We attribute this phenomenon to the complexity of urban land covers. By masking HSI patches with a higher ratio, the diversity of feature representations could be enhanced, thereby preventing the model from overfitting to specific HSI features. Consequently, we set this parameter to 0.75 for subsequent experiments.

7) Learning Rate During the Fine-Tuning Stage: The learning rate during the fine-tuning stage is set to [5e-4, 8e-4, 1e-3, 3e-3, 5e-3] for fine-tuning, and the experimental results are shown in Fig. 3(g). Based on the results, this parameter is set to 1e-3 for PU, SA, and LK datasets, and 3e-3 for HU datasets in the subsequent experiments.

D. Ablation Study

1) Conditional Transformer: In this part of the experiment, we compare the classification accuracy achieved by employing standard or conditional Transformer blocks to construct the encoder or decoder of DEMAE. As is shown in Table II, using conditional Transformer blocks to construct the encoder

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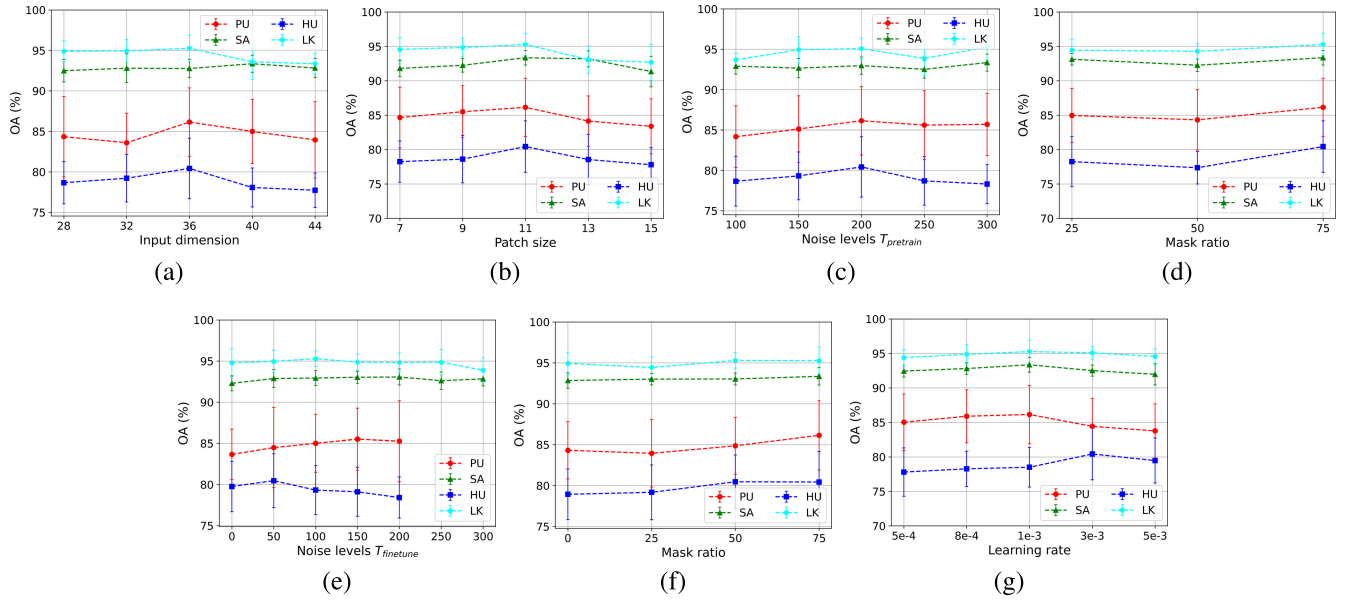


Fig. 3. Evaluation of OA as a function of (a) input dimension, (b) patch size, (c) noise levels $T_{\text{pre-train}}$, (d) mask ratio during the pre-training stage and (e) noise levels $T_{\text{fine-tune}}$, (f) mask ratio, and (g) learning rate during the fine-tuning stage.

TABLE II
ABLATION STUDY OF CONDITIONAL TRANSFORMER FOR FOUR DATASETS

Datasets	Encoder	Decoder	OA	AA	$\kappa\times100$
PU	*Std.*†	*Std.*	84.00±4.80	86.24±3.05	79.46±5.76
	Std.	*Cond.*‡	83.83±3.90	86.39±2.73	79.26±4.63
	Cond.	*Std.*	**86.14±4.23**	87.31±2.34	**82.14±5.02**
	Cond.	*Cond.*	85.28±4.59	**87.39±2.16**	81.10±5.46
SA	*Std.*	*Std.*	91.85±1.86	95.54±1.14	90.94±2.04
	Std.	*Cond.*	92.41±1.03	96.02±0.90	91.57±1.14
	Cond.	*Std.*	**93.35±1.06**	**96.39±0.66**	**92.61±1.18**
	Cond.	*Cond.*	92.58±1.38	96.20±0.82	91.76±1.52
HU	*Std.*	*Std.*	76.83±2.96	80.10±2.49	74.99±3.19
	Std.	*Cond.*	78.80±2.92	81.78±2.49	77.11±3.15
	Cond.	*Std.*	**80.43±3.74**	**83.16±3.15**	**78.86±4.03**
	Cond.	*Cond.*	78.38±3.61	81.33±2.98	76.66±3.89
LK	*Std.*	*Std.*	93.61±1.72	92.40±1.41	91.73±2.19
	Std.	*Cond.*	93.54±1.61	93.00±1.57	91.64±2.04
	Cond.	*Std.*	**95.27±1.63**	**93.86±1.65**	**93.86±2.09**
	Cond.	*Cond.*	94.70±0.92	93.32±1.39	93.12±1.17

† "*Std.*" represents the encoder or decoder is constructed using standard Transformer blocks.
‡ "*Cond.*" represents the encoder or decoder is constructed using the proposed conditional Transformer blocks.

achieves higher classification accuracy. On this basis, using standard Transformer blocks for the decoder results in the best classification performance among all model variants. In addition, the accuracy improvement on HU dataset is more notable. We believe that conditional Transformer block empowers the model to effectively extract invariant features from unlabeled mask samples with varying scales of noise during the pre-training stage and integrate this prior knowledge into subsequent classification tasks so as to guide the model to quickly acquire discriminative feature extraction capabilities. As a result, this process contributes to the enhancement of classification accuracy.

2) Weights Initialization Methods for Encoder: The core of DEMAE is its weights initialization strategy for the encoder, which involves self-supervised pre-training utilizing diffusion and MAE branches for denoising and reconstruction tasks. To verify their effectiveness, this part of the experiment compares the classification accuracy achieved by the encoder of DEMAE under various initialization methods, including random initialization, sole utilization of the diffusion or MAE branch for initialization, and simultaneous utilization of both branches for initialization. Specifically, when only the diffusion branch is used for initialization, the HSI patch will not be masked throughout the process. Conversely, when only the MAE branch is used, the HSI patch will not be noisy.

The experimental results illustrated in Table III demonstrate DEMAE's superior classification accuracy on SA and LK datasets. While the model variant solely employing MAE or diffusion branch outperforms DEMAE on PU and HU datasets, the gap is negligible. We believe that the shadows present in HU dataset can be regarded as regions characterized by extensive noise, where the reconstruction task of MAE branch impedes the feature learning process of the overall HSI, while the denoising task of diffusion branch is particularly suitable for feature mining within such areas. On the contrary, PU dataset exhibits higher data quality, with significant spectral differences between land covers. Employing MAE branch for pre-training serves to prevent the model from overfitting to specific HSI feature representation so as to obtain satisfactory initial weights. However, the efficacy of diffusion branch in enhancing performance is less obvious.

In summary, regardless of whether employing a single branch or dual branches for self-supervised pre-training, the classification performance can be improved compared with random initialization method.

3) Loss Function for the Pre-Training Stage: In this part of the experiment, we compare the OA obtained by two loss function algorithms during the pre-training stage, where the output of the diffusion branch predicts original image x_0^{vis} and added noise ϵ, respectively. Their formulas are presented

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TABLE III
ABLATION STUDY OF WEIGHTS INITIALIZATION METHODS FOR FOUR DATASETS

Datasets	Initialization	OA	AA	$\kappa\times100$
PU	Random†	77.85±5.42	82.23±2.06	71.81±6.08
	Only MAE‡	**86.42±3.21**	86.78±3.52	**82.34±4.02**
	Only Diffusion£	84.25±3.45	87.08±2.02	79.70±4.00
	DEMAEß	86.14±4.23	**87.31±2.34**	82.14±5.02
SA	Random	91.69±0.81	91.69±0.81	91.69±0.81
	Only MAE	91.85±1.76	95.57±1.27	90.95±1.95
	Only Diffusion	92.07±0.85	94.98±0.66	91.18±0.95
	DEMAE	**93.35±1.06**	**96.39±0.66**	**92.61±1.18**
HU	Random	77.05±2.95	77.05±2.95	77.05±2.95
	Only MAE	78.30±1.82	81.35±1.50	76.57±1.96
	Only Diffusion	**80.73±2.41**	**83.63±2.08**	**79.19±2.60**
	DEMAE	80.43±3.74	83.16±3.15	78.86±4.03
LK	Random	90.91±1.83	90.91±1.83	90.91±1.83
	Only MAE	91.73±2.26	90.83±1.26	89.36±2.80
	Only Diffusion	93.86±1.19	91.66±1.54	92.03±1.52
	DEMAE	**95.27±1.63**	**93.86±1.65**	**93.86±2.09**

† "Random" represents the weights of encoder are initialized using random initialization.
‡ "Only MAE" represents only the MAE branch is used to initialize the weights of encoder.
£ "Only Diffusion" represents only the diffusion branch is used to initialize the weights of encoder.
ß "DEMAE" represents both the MAE and diffusion branches are used to initialize the weights of encoder.

TABLE IV
ABLATION STUDY OF LOSS FUNCTION DURING PRE-TRAINING STAGE FOR FOUR DATASETS

Datasets	Loss function	OA	AA	$\kappa\times100$
PU	Eq.(23)	**86.14±4.23**	87.31±2.34	**82.14±5.02**
	Eq.(25)	85.59±3.53	**87.82±2.74**	81.46±4.22
SA	Eq.(23)	**93.35±1.06**	**96.39±0.66**	**92.61±1.18**
	Eq.(25)	92.07±1.43	95.77±0.94	91.20±1.57
HU	Eq.(23)	**80.43±3.74**	**83.16±3.15**	**78.86±4.03**
	Eq.(25)	78.16±2.87	81.19±2.58	76.42±3.10
LK	Eq.(23)	**95.27±1.63**	**93.86±1.65**	**93.86±2.09**
	Eq.(25)	93.86±1.34	92.30±1.45	92.03±1.69

in (23) and (25). Among them, (23) is the loss function employed by DEMAE during the self-supervised pre-training stage, while the loss function for the diffusion branch in (25) is consistent with that of conventional DDPM [27]

$$L_{\text{pre-train}} = L_{\text{Diffusion}} + L_{\text{MAE}} = \left\|\hat{\epsilon}_\theta\left(x_t^{\text{vis}}, t\right) - \epsilon\right\|_2^2 + \left\|\hat{x}_\theta^{\text{mask}} - x^{\text{mask}}\right\|_2^2 \quad (25)$$

where $\hat{\epsilon}_\theta(x_t^{\text{vis}}, t)$ represents the predicted noise output by the diffusion branch, and ϵ denotes the actual noise component added to the masked HSI patch during the HSI data preprocessing stage. Additionally, the meanings of other symbols are same with that in (23).

The experimental results are shown in Table IV. Using (23) to calculate the loss of the diffusion branch achieves higher classification accuracy across almost all indicators on four datasets. Additionally, we found that the loss of the pre-trained model at convergence is lower when using (23). Therefore, we believe that (23) is more conducive for the model to learn discriminative features of HSI data.

4) Loss Function for the Fine-Tuning Stage: The proposed SNR-Enhanced loss function introduces the weighted ratio of data and noise from samples into the modulating factor γ of the traditional focal loss, aiming to standardize the model training process. To validate its effectiveness, we compare the classification accuracy achieved by utilizing the SNR-Enhanced loss and focal loss during training. We set the parameter γ to [0.5, 1, 2], respectively, noting that the focal loss is equivalent to cross-entropy loss when γ is set to 1. The experimental results detailed in Table V demonstrate that employing the SNR-Enhanced loss function for training can achieve higher classification accuracy across the four datasets.

TABLE V
ABLATION STUDY OF LOSS FUNCTION DURING FINE-TUNING STAGE FOR FOUR DATASETS

Datasets	Loss function	OA	AA	$\kappa\times100$
PU	Cross Entropy†	84.49±3.89	86.73±1.44	80.09±4.49
	Focal (γ=0.5)‡	83.94±5.00	86.39±2.32	79.48±5.89
	Focal (γ=2)£	84.89±3.15	86.90±1.14	80.54±3.70
	SNR-Enhancedß	**86.14±4.23**	**87.31±2.34**	**82.14±5.02**
SA	Cross Entropy	92.72±1.35	95.99±0.84	91.91±1.49
	Focal (γ=0.5)	92.42±1.04	95.80±0.70	91.58±1.14
	Focal (γ=2)	92.65±1.19	96.05±0.86	91.84±1.31
	SNR-Enhanced	**93.35±1.06**	**96.39±0.66**	**92.61±1.18**
HU	Cross Entropy	79.26±2.95	82.27±2.45	77.60±3.17
	Focal (γ=0.5)	78.77±3.12	81.79±2.44	77.08±3.36
	Focal (γ=2)	78.71±3.27	81.85±2.77	77.01±3.52
	SNR-Enhanced	**80.43±3.74**	**83.16±3.15**	**78.86±4.03**
LK	Cross Entropy	94.79±1.38	93.64±1.32	93.24±1.76
	Focal (γ=0.5)	94.57±1.31	93.61±1.14	92.96±1.67
	Focal (γ=2)	93.91±1.83	93.45±1.28	92.09±2.33
	SNR-Enhanced	**95.27±1.63**	**93.86±1.65**	**93.86±2.09**

† "Cross Entropy" represents the encoder is trained using cross entropy loss in the fine-tuning stage.
‡ "Focal (γ=0.5)" represents the encoder is trained using focal loss with γ set to 0.5 in the fine-tuning stage.
£ "Focal (γ=2)" represents the encoder is trained using focal loss with γ set to 2 in the fine-tuning stage.
ß "SNR-Enhanced" represents the encoder is trained using the proposed SNR-Enhanced loss in the fine-tuning stage.

E. Comparison of Different Methods With Different Numbers of Labeled Samples

In this part of the experiment, we compare the experimental accuracy achieved by different models under different numbers of labeled samples to evaluate the generalization performance of DEMAE. As is shown in Fig. 4, the proposed DEMAE outperforms other models across PU, HU, and LK datasets, with particularly notable advantages on PU and HU datasets. On SA dataset, the classification accuracy of SpectralDiff surpasses that of DEMAE as the number of labeled samples increases. The explanation we give is that this diffusion-based method stacks diffusion features of samples with diverse noise scales, expanding the feature dimension to enhance feature mining capabilities when numbers of labeled samples are sufficient. In contrast, the feature dimension output by DEMAE does not exceed 100, offering a more computationally efficient solution. Overall, DEMAE achieves superior classification performance in most cases, which can prove its strong generalization capability.

F. Classification Results

To validate the superiority of DEMAE for HSI classification with a limited number of labeled samples, we compare the

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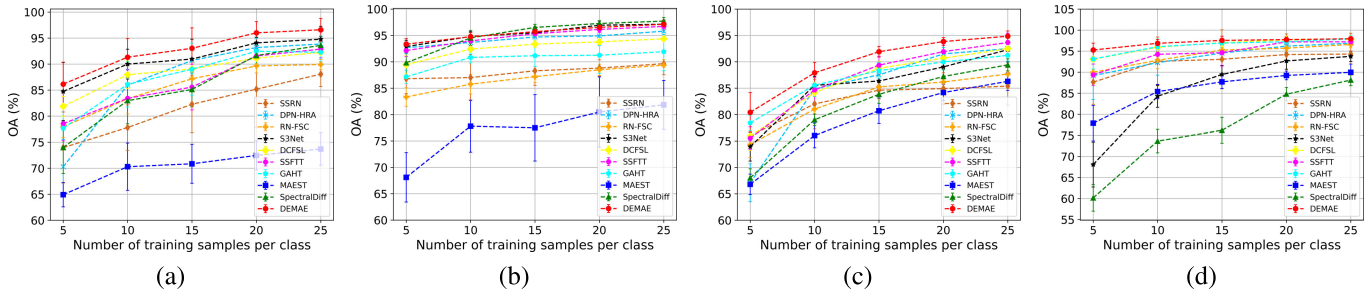


Fig. 4. Evaluation of OA as a function of training samples per class. (a) PU. (b) SA. (c) HU. (d) LK.

classification accuracy achieved by different models using five labeled samples per class on PU, SA, HU, and LK datasets. The classification accuracy results for each model are summarized in Tables VI– IX, and the corresponding classification maps are visualized in Figs. 5–8.

1) PU Dataset: The classification accuracy results for PU dataset, presented in Table VI, demonstrate DEMAE's superior performance compared with other methods under OA and $\kappa \times 100$, with improvements ranging from 1.41% to 21.23% and 1.58% to 26.96%, respectively. As for AA, where S3Net attains higher accuracy, we believe that the disparity between DEMAE and S3Net is mainly evident in the classification of gravel and self-blocking bricks. The contrastive learning-based S3Net exhibits an advantage in distinguishing between similar spectral land covers. In addition, according to Fig. 5, DEMAE yields a smoother classification effect, especially in the bare soil region indicated in yellow.

2) SA Dataset: The classification accuracy results obtained by different methods on SA dataset are shown in Table VII. DEMAE achieves the highest accuracy under OA, AA, and $\kappa \times 100$. Compared with other methods, DEMAE exhibits accuracy improvements ranging from 0.46% to 25.25%, 0.83% to 21.49% and 0.51% to 27.85%, respectively. Additionally, DEMAE attains the highest accuracy in 6 out of 16 classes. According to the classification maps presented in Fig. 6, DEMAE achieves a more uniform classification effect between fields, resulting in reduced presence of impurities within the fields.

3) HU Dataset: The classification accuracy results for HU dataset, presented in Table VIII, demonstrate that DEMAE achieves at least 2.00%, 1.52%, and 2.16% improvements compared with other methods under OA, AA, and $\kappa \times 100$, respectively. The classification maps generated by different methods are shown in Fig. 7. DEMAE distinguishes the upper half shadow region more effectively, producing finer classification results compared with most comparison models. We attribute this phenomenon to DEMAE's capability to learn distinct features for shadow and non-shadow areas during the self-supervised pre-training stage. As a result, DEMAE will not roughly identify large shadow areas as same class during the classification stage.

4) LK Dataset: The classification accuracy results for LK dataset are presented in Table IX. Compared with other methods, DEMAE achieves the highest accuracy across OA, AA, and $\kappa \times 100$, and the improvement rate is at least 2.14%. In addition, DEMAE attains the highest accuracy in six out of nine classes. The classification maps generated by different methods are illustrated in Fig. 8. Notably, the classification map produced by DEMAE exhibits a smoother appearance and contains fewer fragments inside the same land cover area.

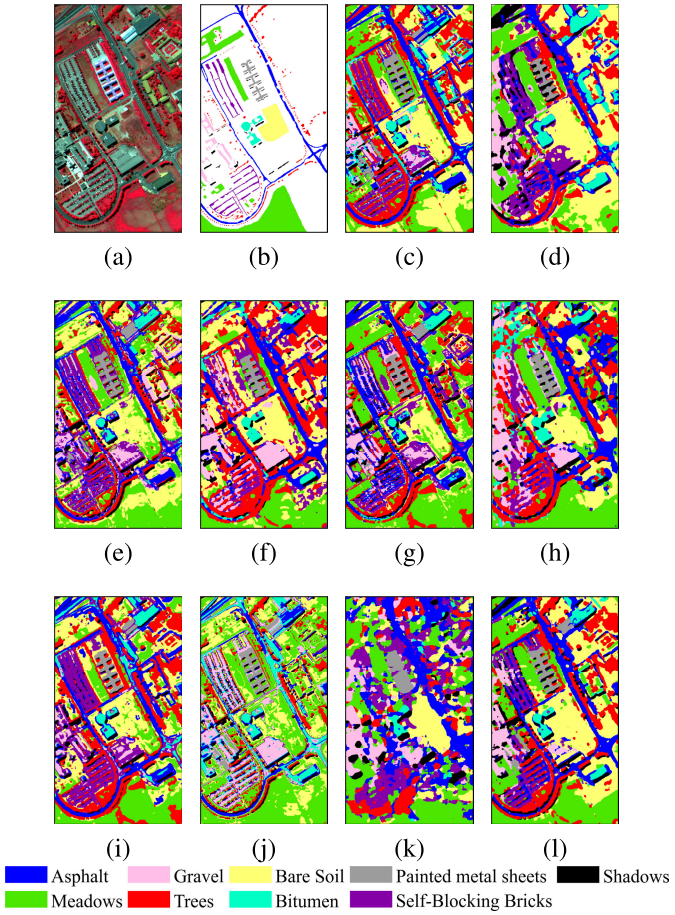


Fig. 5. Classification maps obtained by different methods for PU dataset (five labeled samples per class for training). (a) False-color composite image (R: 102, G: 56, B: 31). (b) Ground truth. (c) SSRN (73.96%). (d) DPN-HRA (70.27%). (e) RN-FSC (77.95%). (f) S3Net (84.73%). (g) DCFSL (81.86%). (h) SSFTT (78.53%). (i) GAHT (77.66%). (j) MAEST (64.91%). (k) SpectralDiff (73.98%). (l) DEMAE (86.14%).

IV. Discussion

In this section, we examine the feature separability of various DEMAE model variants and perform cross-domain experiments on different datasets. In addition, we explore the relationship between the proposed DEMAE and existing interpretability models.

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TABLE VI
CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR PU DATASET (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	SSRN [6]	DPN-HRA [46]	RN-FSC [47]	S3Net [48]	DCFSL [49]	SSFTT [13]	GAHT [15]	MAEST [22]	SpectralDiff [44]	DEMAE (Ours)
1	85.71±5.50	**91.84±5.35**	72.78±12.97	85.99±7.48	77.89±12.85	63.68±5.85	84.16±8.79	57.68±9.04	65.17±4.60	77.01±7.73
2	61.21±17.80	**94.54±4.09**	83.44±4.30	81.82±14.35	87.38±3.86	81.52±9.57	73.16±13.48	71.87±7.52	70.76±10.33	88.31±10.67
3	65.72±14.23	44.87±12.66	67.36±11.15	**82.82±7.02**	63.53±12.97	78.87±8.68	64.47±9.24	47.32±15.80	81.83±7.47	75.69±13.22
4	91.22±5.27	63.97±16.20	91.00±3.89	89.41±4.48	92.50±4.36	84.05±6.92	**93.98±5.31**	84.19±7.31	60.11±8.03	91.20±4.37
5	96.87±5.46	98.29±1.90	99.44±0.53	99.27±1.07	99.31±0.98	99.70±0.45	99.74±0.16	90.75±10.25	91.19±13.84	**99.75±0.46**
6	77.10±13.20	65.75±21.53	57.86±5.41	84.79±10.31	74.07±5.47	80.27±12.91	70.74±13.10	45.89±7.25	85.57±15.11	**90.55±7.26**
7	93.21±6.04	42.61±23.60	68.94±8.79	**99.77±0.40**	78.94±7.90	99.34±0.63	84.39±7.79	66.42±9.52	79.35±18.38	96.02±3.35
8	81.56±8.79	58.27±12.19	72.20±10.04	84.29±18.06	63.53±11.79	65.09±8.37	76.23±11.39	47.28±12.31	**88.61±7.11**	78.00±12.23
9	**99.71±0.60**	79.93±15.11	98.45±1.64	82.24±10.76	99.11±0.76	88.83±9.30	99.47±0.60	86.07±10.44	76.19±9.53	89.24±8.41
OA (%)	73.96±6.82	70.27±5.23	77.95±2.05	84.73±5.57	81.86±1.91	78.53±3.26	77.66±4.11	64.91±2.34	73.98±4.99	**86.14±4.23**
AA (%)	83.59±2.82	71.12±3.09	79.05±1.00	**87.82±2.95**	81.81±1.47	82.37±2.77	82.93±1.32	66.38±1.74	77.64±3.88	87.31±2.34
$\kappa\times100$	67.89±7.31	62.82±5.90	71.40±2.38	80.56±6.45	76.41±2.31	72.46±3.84	71.74±4.38	55.18±2.49	67.27±5.62	**82.14±5.02**

TABLE VII
CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR SA DATASET (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	SSRN [6]	DPN-HRA [46]	RN-FSC [47]	S3Net [48]	DCFSL [49]	SSFTT [13]	GAHT [15]	MAEST [22]	SpectralDiff [44]	DEMAE (Ours)
1	99.57±0.70	99.52±0.87	96.26±2.47	98.40±2.98	98.99±1.12	98.62±2.42	**99.85±0.37**	94.81±4.07	88.32±9.10	99.43±1.53
2	99.61±0.49	98.59±1.30	98.71±1.09	97.13±5.71	99.59±0.92	99.23±2.10	97.95±2.28	89.64±7.89	91.91±4.02	**99.99±0.02**
3	89.10±6.81	96.11±5.14	82.51±5.25	97.98±5.08	90.44±10.53	98.19±4.94	85.26±8.05	52.58±18.35	96.63±6.92	**99.93±0.14**
4	99.45±1.08	89.54±6.68	98.77±1.28	98.91±2.08	**99.48±0.62**	98.45±1.96	99.29±1.15	99.00±0.98	98.88±1.83	99.36±0.98
5	91.11±10.80	95.81±3.16	90.63±6.78	**97.36±2.74**	91.08±3.03	93.18±5.57	91.55±3.87	75.74±15.18	90.79±5.94	91.05±6.77
6	99.57±0.68	99.32±0.68	98.73±0.97	99.10±1.05	98.44±2.20	98.60±2.00	**99.99±0.02**	94.38±3.17	82.32±7.97	97.80±2.80
7	99.80±0.25	98.41±2.57	93.85±4.49	**99.92±0.20**	98.98±1.77	99.41±0.81	99.71±0.36	73.72±25.26	92.80±8.07	99.59±0.85
8	66.02±15.88	**92.73±4.12**	68.69±9.68	80.37±9.15	77.86±9.76	80.73±5.80	65.99±11.54	60.48±11.90	79.52±9.51	82.30±4.96
9	96.92±3.07	99.00±0.92	91.43±1.46	99.45±1.19	99.07±1.26	99.86±0.27	98.09±2.25	65.35±30.92	95.22±3.76	**100.00±0.00**
10	84.76±6.13	94.39±4.45	71.12±6.56	91.35±5.73	87.02±3.49	92.26±4.24	86.48±5.80	41.48±16.71	91.91±4.65	**94.39±3.04**
11	96.99±5.52	96.47±5.96	82.81±15.07	99.32±1.04	98.57±1.36	99.79±0.21	97.41±2.82	59.33±17.41	97.23±2.46	**99.81±0.36**
12	98.55±2.55	91.79±6.14	93.13±3.48	91.10±11.22	**99.61±0.56**	97.64±3.30	99.16±1.16	70.50±21.87	96.89±2.70	98.37±2.23
13	99.20±1.22	86.33±18.63	96.02±2.72	97.52±3.40	**99.25±0.34**	97.17±4.00	98.50±2.95	97.15±2.52	97.73±3.78	98.36±2.31
14	97.90±1.68	88.56±16.30	97.91±3.27	94.74±6.07	**98.55±1.73**	89.91±10.54	97.89±1.12	92.68±6.27	90.70±5.25	95.55±6.60
15	75.89±7.83	78.56±6.25	68.20±6.21	90.65±14.73	73.04±6.60	84.00±6.97	77.21±9.93	42.48±21.66	**92.88±5.85**	86.95±7.06
16	85.02±7.20	96.68±7.07	82.64±10.00	95.70±4.61	91.71±6.05	94.79±4.08	88.91±7.16	89.05±7.10	95.05±5.03	**99.40±0.72**
OA (%)	86.80±2.82	92.69±1.10	83.34±1.75	92.89±1.19	89.46±1.68	92.15±1.11	87.15±1.24	68.10±4.68	89.82±2.31	**93.35±1.06**
AA (%)	92.47±1.54	93.86±2.26	88.21±1.49	95.56±0.91	93.86±0.76	95.11±1.17	92.70±0.65	74.90±5.54	92.42±1.59	**96.39±0.66**
$\kappa\times100$	85.37±3.08	91.88±1.21	81.50±1.93	92.10±1.33	88.29±1.84	91.28±1.22	85.76±1.34	64.76±5.15	88.69±2.55	**92.61±1.18**

TABLE VIII
CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR HU DATASET (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	SSRN [6]	DPN-HRA [46]	RN-FSC [47]	S3Net [48]	DCFSL [49]	SSFTT [13]	GAHT [15]	MAEST [22]	SpectralDiff [44]	DEMAE (Ours)
1	86.57±9.17	86.52±7.70	85.69±7.33	74.40±11.38	88.92±6.40	73.73±12.00	**90.26±6.51**	90.21±6.76	68.88±12.16	81.56±14.07
2	80.94±7.93	77.25±15.13	84.88±6.66	86.58±6.10	79.03±11.23	85.36±7.90	83.51±9.90	74.03±10.06	72.77±9.91	**87.85±6.30**
3	**99.62±0.95**	98.67±2.24	86.01±7.12	99.35±0.98	96.53±3.19	96.49±2.45	94.61±8.24	88.51±8.42	92.77±7.36	98.53±1.66
4	92.08±3.53	81.81±13.16	**94.96±2.28**	91.34±5.78	90.79±3.21	82.21±10.02	94.77±2.22	84.89±2.96	67.02±12.36	90.24±4.13
5	93.51±9.01	89.28±8.74	90.49±11.72	88.24±7.92	95.24±3.80	94.02±7.78	98.56±1.65	91.68±7.91	95.97±5.72	**98.58±3.66**
6	87.91±7.00	**94.45±5.35**	77.94±5.67	90.00±5.80	80.97±6.58	85.97±7.41	87.62±5.63	61.84±6.22	81.53±4.46	91.19±5.97
7	75.35±9.10	63.49±17.86	72.38±9.24	63.61±8.70	63.06±13.90	64.97±10.48	**76.96±5.25**	54.09±12.61	67.80±7.75	74.70±11.04
8	34.88±9.66	**66.60±20.55**	39.59±19.77	41.98±7.86	44.67±5.01	42.82±4.57	47.94±9.50	39.13±9.51	31.44±7.57	51.09±6.44
9	73.24±10.47	44.30±13.42	59.05±9.02	51.11±4.47	63.18±9.36	55.17±16.35	**79.00±3.36**	58.27±14.48	67.46±7.85	73.20±6.49
10	61.31±17.88	47.61±10.97	63.86±17.38	64.79±18.13	66.04±14.39	**75.14±14.98**	55.87±10.96	55.92±7.99	51.32±10.21	75.09±18.17
11	66.22±9.92	62.02±23.41	56.19±9.69	79.18±10.03	62.93±8.75	76.86±13.44	65.20±5.35	54.42±8.48	54.18±9.61	**82.93±12.43**
12	52.93±11.99	55.81±15.07	60.78±16.47	50.73±10.76	**73.66±11.54**	65.22±11.01	55.12±12.25	47.17±9.92	54.86±8.81	57.60±10.32
13	90.04±4.29	54.87±18.20	95.60±1.47	90.91±3.36	86.66±7.66	90.88±5.72	**95.73±0.50**	29.59±10.65	88.36±4.64	87.00±8.83
14	99.79±0.56	95.29±5.21	96.45±3.91	**100.00±0.00**	88.30±8.13	99.57±0.91	99.49±0.54	87.90±4.50	99.98±0.07	**100.00±0.00**
15	**100.00±0.00**	85.26±8.17	97.15±2.50	99.85±0.46	95.73±2.57	97.85±5.01	99.99±0.02	92.89±2.01	91.56±5.63	97.91±3.54
OA (%)	75.99±2.33	67.09±3.58	74.33±2.44	73.93±2.72	75.91±2.67	75.54±2.14	78.43±1.45	66.80±1.94	68.04±1.70	**80.43±3.74**
AA (%)	79.63±1.68	73.55±3.53	77.40±1.81	78.14±2.22	78.38±2.41	79.08±1.68	81.64±1.39	67.37±1.46	72.39±1.47	**83.16±3.15**
$\kappa\times100$	74.11±2.50	64.49±3.85	72.28±2.63	71.87±2.93	73.97±2.88	73.58±2.31	76.70±1.57	64.15±2.08	65.47±1.85	**78.86±4.03**

A. Feature Separability

We evaluate the feature separability of labeled samples obtained through PCA whitening and feature extraction using either pre-trained or fine-tuned models across four datasets from both qualitative and quantitative perspectives.

1) For Qualitative Analysis: The t-SNE is used to map the features of labeled HSI samples onto a 2-D plane, where the points are marked according to their category color. The feature separability graphs of different datasets are shown in Fig. 9, where (a)–(c), (d)–(f), (g)–(i), and (j)–(l) correspond to PU, SA, HU, and LK, respectively. It is evident that the separability of PCA whitening features is poor, whereas features extracted using the pre-trained model exhibit a significant degree of separability. It should be emphasized that training a self-supervised pre-trained model does not involve any label information. Instead, it employs a heuristic training strategy that allows the model to autonomously identify key information in HSI data, verifying the feasibility of the proposed DEMAE for HSI feature mining. In addition, the feature separability can be further improved by using the fine-tuned model, which is trained with five labeled samples per class. This approach has the most significant effect on the HU dataset.

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TABLE IX
CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR LK DATASET (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	SSRN [6]	DPN-HRA [46]	RN-FSC [47]	S3Net [48]	DCFSL [49]	SSFTT [13]	GAHT [15]	MAEST [22]	SpectralDiff [44]	DEMAE (Ours)
1	93.50±9.70	96.49±4.03	96.66±2.05	76.78±14.82	96.91±3.34	96.35±3.96	95.42±4.95	83.10±9.65	59.54±9.27	**98.71±1.63**
2	73.47±18.79	74.66±13.13	87.22±6.48	95.27±6.02	86.63±9.26	87.60±9.34	84.54±14.22	40.99±9.98	77.72±10.89	**95.59±3.09**
3	70.81±21.70	59.45±23.71	88.60±3.35	98.64±1.25	91.01±6.51	98.69±1.97	92.31±7.45	74.04±10.59	98.14±1.63	**99.28±1.71**
4	77.09±13.80	**97.24±2.46**	80.50±8.01	73.45±10.50	88.20±7.56	78.27±6.23	87.42±6.60	61.15±14.85	53.49±12.21	94.39±3.52
5	85.84±12.67	66.22±22.22	91.55±4.81	96.65±8.46	92.15±8.99	93.82±9.07	92.13±5.21	58.65±14.66	92.48±7.49	**98.13±3.43**
6	93.59±5.97	84.66±10.35	90.90±7.97	82.68±8.38	92.81±3.13	92.67±3.89	94.04±3.87	83.17±5.32	66.07±8.55	**96.43±2.69**
7	96.66±8.37	99.38±0.88	**99.96±0.04**	52.35±11.79	99.83±0.13	96.92±1.47	99.69±0.21	99.65±0.16	63.22±10.24	96.10±2.23
8	87.61±10.16	82.31±9.05	71.13±12.26	52.37±12.63	79.92±12.29	78.04±8.47	**90.21±7.71**	67.29±12.99	34.62±7.24	79.02±13.22
9	80.32±15.90	51.27±10.26	60.18±14.48	51.90±7.63	70.43±10.36	83.02±8.64	79.60±12.14	45.45±10.06	50.89±9.29	**87.08±6.46**
OA (%)	87.62±5.27	89.17±5.67	89.98±2.51	68.08±5.33	93.07±2.33	89.38±2.55	93.13±2.37	77.88±4.21	60.15±3.14	**95.27±1.63**
AA (%)	84.32±3.90	79.08±3.76	85.19±2.81	75.57±2.88	88.66±2.97	89.49±2.98	90.60±2.59	68.17±2.51	66.24±1.80	**93.86±1.65**
$\kappa\times100$	84.27±6.39	86.23±6.79	87.12±3.11	60.77±5.86	91.00±2.95	86.36±3.23	91.09±3.01	72.15±4.88	48.50±3.18	**93.86±2.09**

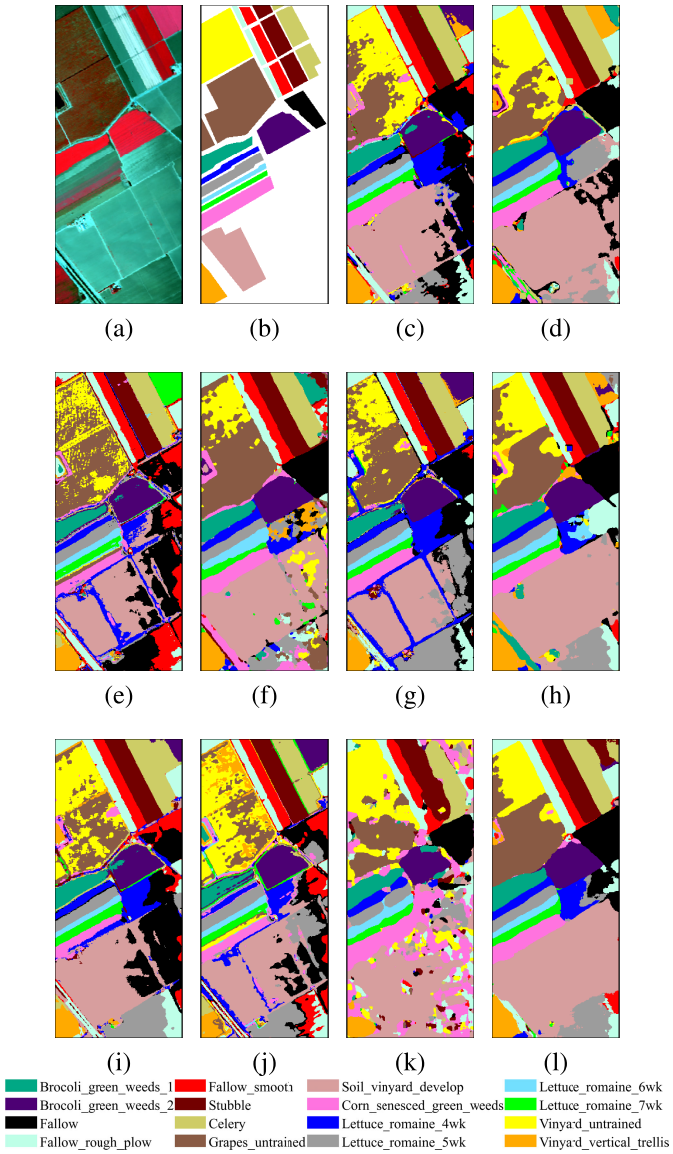


Fig. 6. Classification maps obtained by different methods for SA dataset (five labeled samples per class for training). (a) False-color composite image (R: 57, G: 27, B: 17). (b) Ground truth. (c) SSRN (86.80%). (d) DPN-HRA (92.69%). (e) RN-FSC (83.34%). (f) S3Net (92.89%). (g) DCFSL (89.46%). (h) SSFTT (92.15%). (i) GAHT (87.15%). (j) MAEST (68.10%). (k) SpectralDiff (89.82%). (l) DEMAE (93.35%).

2) For Quantitative Analysis: We use the Fisher discriminant analysis method to calculate the inter-class dispersion of the above three features. The formula is provided in (26), and the corresponding results are detailed in the caption of Fig. 9

$$S_b = \frac{\sum_{i=1,j=1,i\neq j}^{c} (m_i - m_j)(m_i - m_j)^T}{c-1} \tag{26}$$

where m_i and m_j represent the mean features of the ith and jth class, respectively, and c represents the number of categories in the corresponding dataset. It is evident that the inter-class dispersion of features extracted using a fine-tuned model is greater than that of the pre-trained model, and the inter-class dispersion of PCA whitening features is the lowest.

B. Cross-Domain Experiments

In this part of the experiment, we perform self-supervised pre-training approach of DEMAE to the source-domain data, followed by utilizing the pre-trained encoder on the target-domain data for fine-tuning and classification. Throughout this cross-domain experiment, the parameter settings during the pre-training and fine-tuning stages are kept consistent with the corresponding optimal parameters for the source-domain and target-domain data. The experimental results are reported in Table X. The highest classification accuracy can be achieved when the source-domain data and target-domain data are the same, except for the HU dataset. We believe this discrepancy may be due to the scarcity of labeled samples relative to the total number of pixels in the HU dataset, which limits the ability to fully verify the generalization capability of DEMAE on this data. Consequently, the classification accuracy obtained by directly pre-training on HU is slightly lower. Additionally, it is found that cross-domain experiments using HSI of similar scenes can also achieve great classification performance, such as PU and HU datasets for urban scenes and SA and LK datasets for farmland scenes. The accuracy results even surpass those of most comparison methods used in this article.

In summary, cross-domain experiments achieve acceptable results, suggesting the feasibility of heuristic feature learning on source-domain data via self-supervised pre-training. This approach allows for the transfer of source-domain knowledge to target-domain data, thereby improving the classification performance.

C. DEMAE Versus Interpretable DL Models

Existing research on interpretable DL models for HSI processing and analysis mainly employs unfolding [50], [51]

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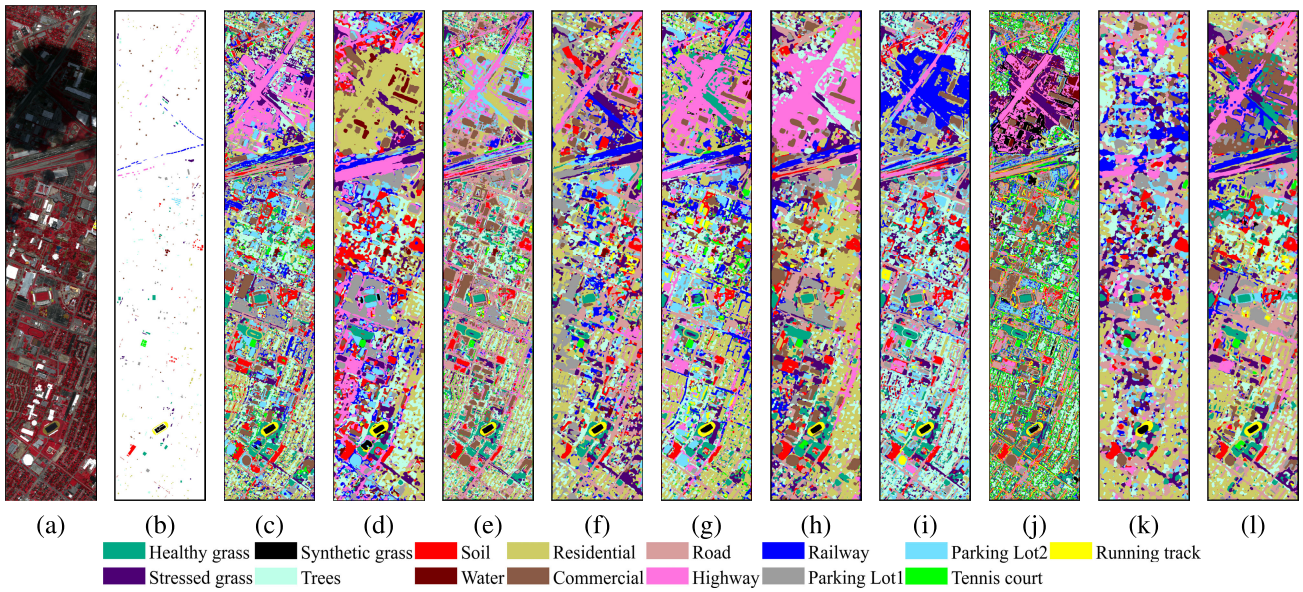


Fig. 7. Classification maps obtained by different methods for HU dataset (five labeled samples per class for training). (a) False-color composite image (R: 59, G: 40, B: 23). (b) Ground Truth. (c) SSRN (75.99%). (d) DPN-HRA (67.09%). (e) RN-FSC (74.33%). (f) S3Net (73.93%). (g) DCFSL (75.91%). (h) SSFTT (75.54%). (i) GAHT (78.43%). (j) MAEST (66.80%). (k) SpectralDiff (68.04%). (l) DEMAE (80.43%).

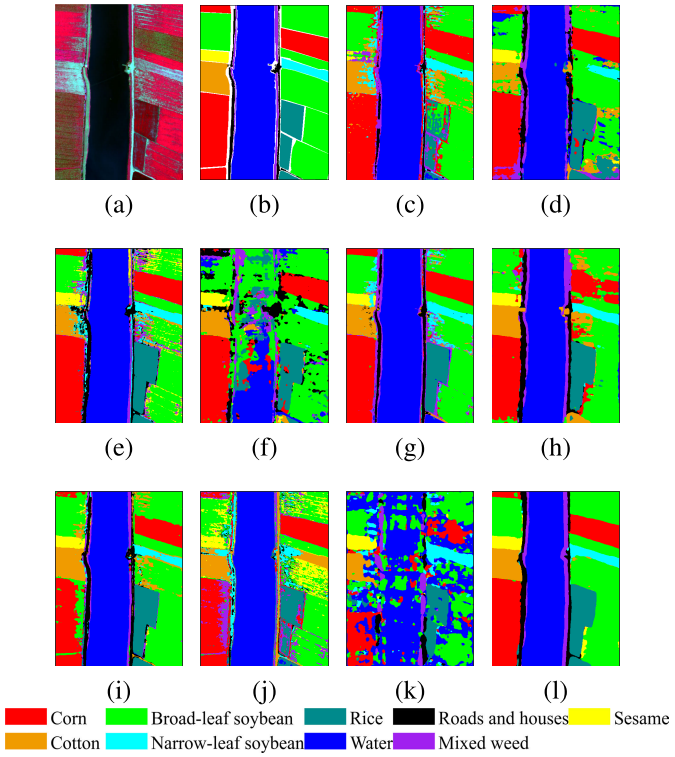


Fig. 8. Classification maps obtained by different methods for LK dataset (five labeled samples per class for training). (a) False-color composite image (R: 180, G: 126, B: 72). (b) Ground Truth. (c) SSRN (87.62%). (d) DPN-HRA (89.17%). (e) RN-FSC (89.98%). (f) S3Net (68.08%). (g) DCFSL (93.07%). (h) SSFTT (89.38%). (i) GAHT (93.13%). (j) MAEST (77.88%). (k) SpectralDiff (60.15%). (l) DEMAE (95.27%).

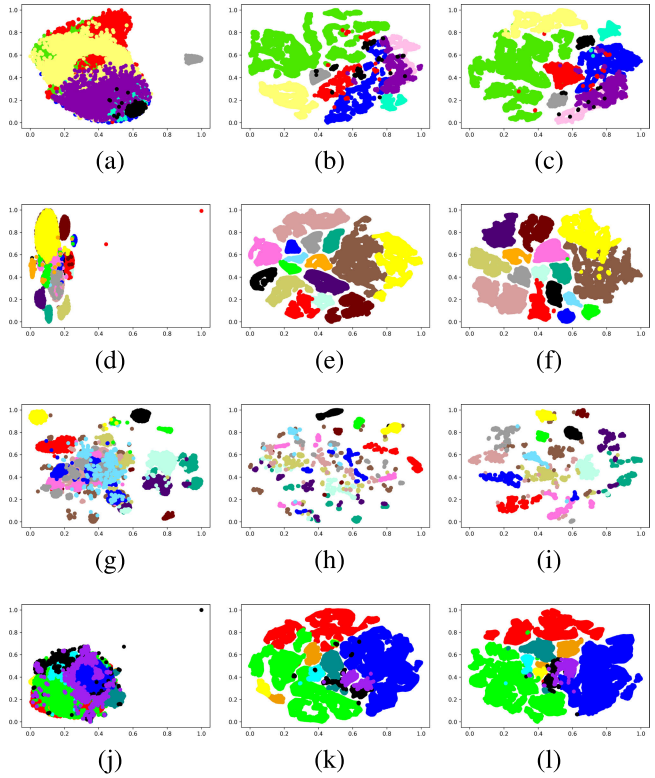


Fig. 9. Feature separability for different model variants (five labeled samples per class for fine-tuning). (a) PCA whitening ($S_b = 0.34$). (b) Pre-trained model ($S_b = 0.97$). (c) Fine-tuned model ($S_b = 1.64$) for PU dataset. (d) PCA whitening ($S_b = 0.49$). (e) Pre-trained model ($S_b = 1.03$). (f) Fine-tuned model ($S_b = 1.54$) for SA dataset. (g) PCA whitening ($S_b = 0.57$). (h) Pre-trained model ($S_b = 0.74$). (i) Fine-tuned model ($S_b = 1.13$) for HU datasets. (j) PCA whitening ($S_b = 0.42$). (k) Pre-trained model ($S_b = 0.73$). (l) Fine-tuned model ($S_b = 2.10$) for LK dataset.

or unrolling [52] techniques, which utilize an iterative optimization process to derive model structure or target task. The proposed DEMAE is similar in that it also employs a series of derivations to determine the target loss function for the training model. However, DEMAE differs by starting from HSI data distribution. Instead of deriving a model structure, it is directly constructed by leveraging the advantages of Transformer.

D. DEMAE Versus RS Foundation Models

Multimodal remote sensing (RS) data can significantly enhance our understanding of the Earth, serving as a critical data basis for training RS foundation models [53]. Therefore,

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TABLE X
CROSS-DOMAIN EXPERIMENTS BETWEEN DIFFERENT DATASETS (FIVE LABELED SAMPLES PER CLASS FOR FINE-TUNING)

Target	Source	OA	AA	$\kappa\times100$
PU	PU	**86.14±4.23**	**87.31±2.34**	**82.14±5.02**
	SA	83.01±4.05	85.95±1.88	78.25±4.72
	HU	84.66±3.37	86.90±2.51	80.23±3.95
	LK	83.85±3.64	87.12±2.12	79.39±4.25
SA	PU	92.88±1.22	96.36±0.94	92.09±1.36
	SA	**93.35±1.06**	96.39±0.66	**92.61±1.18**
	HU	92.80±0.91	95.99±0.64	92.00±1.01
	LK	93.24±0.98	**96.56±0.61**	92.50±1.09
HU	PU	**80.59±2.65**	**83.31±2.24**	**79.05±2.85**
	SA	78.88±3.05	81.97±2.57	77.20±3.29
	HU	80.43±3.74	83.16±3.15	78.86±4.03
	LK	79.01±3.64	82.11±2.96	77.34±3.92
LK	PU	94.33±1.35	91.98±1.47	92.63±1.72
	SA	93.80±1.45	91.93±1.56	91.97±1.85
	HU	93.77±0.93	92.08±1.31	91.93±1.17
	LK	**95.27±1.63**	**93.86±1.65**	**93.86±2.09**

establishing multimodal RS benchmark datasets, such as cross-city [54], is essential for developing foundation models [26] capable of large-scale image information extraction and analysis. At present, DEMAE is limited to HSI feature extraction and classification, handling relatively simple tasks. Future research should focus on utilizing multimodal RS data for training and integrating it with RS foundation models to address more complex challenges.

V. CONCLUSION

In this article, we propose DEMAE, for HSI classification with few labeled samples, which is an MAE-based method comprising primarily of pre-training and fine-tuning stages. During the pre-training stage, masked samples containing differentiated noise levels are employed to stimulate the model to heuristically learn robust features from HSI. During the fine-tuning stage, masked samples with low noise levels are utilized to prevent the pre-trained model from overfitting. Without introducing additional model structures, we simply stack conditional Transformer blocks as encoder and standard Transformer blocks as decoder. Experimental results demonstrate that this straightforward model architecture, along with the proposed SNR-Enhanced loss function, significantly improves classification performance. Moreover, to the best of our knowledge, this study represents an early attempt to apply SSL techniques for cross-domain HSI classification within similar scenarios. In future research, we aim to use HSI from more complex scenes for model pre-training and explore methods to enhance the generalization capabilities of MAE-based and diffusion-based DL models for larger-scale HSI classification tasks. In addition, we are considering ways to make the proposed DEMAE applicable to RS images of different modalities and integrate it with RS foundation models.

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REFERENCES

[1] D. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.
[2] M. A. Moharram and D. M. Sundaram, "Land use and land cover classification with hyperspectral data: A comprehensive review of methods, challenges and future directions," *Neurocomputing*, vol. 536, pp. 90–113, Jun. 2023.
[3] S. Peyghambari and Y. Zhang, "Hyperspectral remote sensing in lithological mapping, mineral exploration, and environmental geology: An updated review," *J. Appl. Remote Sens.*, vol. 15, no. 3, Jul. 2021, Art. no. 031501.
[4] M. Shimoni, R. Haelterman, and C. Perneel, "Hypersectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.
[5] C. Zhao, M. Jia, Z. Wang, D. Mao, and Y. Wang, "Identifying mangroves through knowledge extracted from trained random forest models: An interpretable mangrove mapping approach (IMMA)," *ISPRS J. Photogramm. Remote Sens.*, vol. 201, pp. 209–225, Jul. 2023.
[6] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.
[7] Y. Xu, L. Zhang, B. Du, and F. Zhang, "Spectral–spatial unified networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5893–5909, Oct. 2018.
[8] S. Mei, J. Ji, Y. Geng, Z. Zhang, X. Li, and Q. Du, "Unsupervised spatial–spectral feature learning by 3D convolutional autoencoder for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6808–6820, Sep. 2019.
[9] Z. Niu, G. Zhong, and H. Yu, "A review on the attention mechanism of deep learning," *Neurocomputing*, vol. 452, pp. 48–62, Sep. 2021.
[10] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.
[11] Y. Qing, W. Liu, L. Feng, and W. Gao, "Improved transformer net for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 11, p. 2216, Jun. 2021.
[12] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.
[13] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.
[14] Z. Xue, Q. Xu, and M. Zhang, "Local transformer with spatial partition restore for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4307–4325, 2022.
[15] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.
[16] J. Gui et al., "A survey on self-supervised learning: Algorithms, applications, and future trends," 2023, *arXiv:2301.05712*.
[17] W. Liu et al., "Self-supervised feature learning based on spectral masking for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407715.
[18] D. Hong, J. Yao, C. Li, D. Meng, N. Yokoya, and J. Chanussot, "Decoupled-and-coupled networks: Self-supervised hyperspectral image super-resolution with subpixel fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5527812.
[19] S. Hou, H. Shi, X. Cao, X. Zhang, and L. Jiao, "Hyperspectral imagery classification based on contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521213.

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[20] J. Yue, L. Fang, H. Rahmani, and P. Ghamisi, "Self-supervised learning with adaptive distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501813.

[21] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," 2021, *arXiv:2111.06377*.

[22] D. Ibanez, R. Fernandez-Beltran, F. Pla, and N. Yokoya, "Masked auto-encoding spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542614.

[23] L. Huang, Y. Chen, and X. He, "Spectral–spatial masked transformer with supervised and contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508718.

[24] X. Cao, H. Lin, S. Guo, T. Xiong, and L. Jiao, "Transformer-based masked autoencoder with contrastive loss for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524312.

[25] F. Zhou, C. Xu, G. Yang, R. Hang, and Q. Liu, "Masked spectral–spatial feature prediction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4400913.

[26] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[27] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," 2020, *arXiv:2006.11239*.

[28] K. Preechakul, N. Chatthee, S. Wizadwongsa, and S. Suwajanakorn, "Diffusion autoencoders: Toward a meaningful and decodable representation," 2021, *arXiv:2111.15640*.

[29] W. Xiang, H. Yang, D. Huang, and Y. Wang, "Denoising diffusion autoencoders are unified self-supervised learners," 2023, *arXiv:2303.09769*.

[30] D. A. Hudson et al., "SODA: Bottleneck diffusion models for representation learning," 2023, *arXiv:2311.17901*.

[31] V. Tao Hu, Y. Chen, M. Caron, Y. M. Asano, C. G. M. Snoek, and B. Ommer, "Guided diffusion from self-supervised diffusion features," 2023, *arXiv:2312.08825*.

[32] X. Chen, Z. Liu, S. Xie, and K. He, "Deconstructing denoising diffusion models for self-supervised learning," 2024, *arXiv:2401.14404*.

[33] W. Peebles and S. Xie, "Scalable diffusion models with transformers," 2022, *arXiv:2212.09748*.

[34] F. Bao et al., "All are worth words: A ViT backbone for diffusion models," 2022, *arXiv:2209.12152*.

[35] F. Bao et al., "One transformer fits all distributions in multi-modal diffusion at scale," 2023, *arXiv:2303.06555*.

[36] D. Baranchuk, I. Rubachev, A. Voynov, V. Khrulkov, and A. Babenko, "Label-efficient semantic segmentation with diffusion models," 2021, *arXiv:2112.03126*.

[37] E. B. Asiedu, S. Kornblith, T. Chen, N. Parmar, M. Minderer, and M. Norouzi, "Decoder denoising pretraining for semantic segmentation," 2022, *arXiv:2205.11423*.

[38] Z. Wang, D. Li, M. Zhang, H. Luo, and M. Gong, "Enhancing hyperspectral images via diffusion model and group-autoencoder super-resolution network," 2024, *arXiv:2402.17285*.

[39] Y. Miao, L. Zhang, L. Zhang, and D. Tao, "DDS2M: Self-supervised denoising diffusion spatio-spectral model for hyperspectral image restoration," 2023, *arXiv:2303.06682*.

[40] J. Yue, L. Fang, S. Xia, Y. Deng, and J. Ma, "Dif-fusion: Toward high color fidelity in infrared and visible image fusion with diffusion models," *IEEE Trans. Image Process.*, vol. 32, pp. 5705–5720, 2023.

[41] J. Chen, S. Liu, Z. Zhang, and H. Wang, "Diffusion subspace clustering for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 6517–6530, 2023.

[42] K. Deng, Y. Qian, J. Nie, and J. Zhou, "Diffusion-model-based hyperspectral unmixing using spectral prior distribution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519716.

[43] X. Zhang, S. Tian, G. Wang, H. Zhou, and L. Jiao, "DiffUCD: Unsupervised hyperspectral image change detection with semantic correlation diffusion model," 2023, *arXiv:2305.12410*.

[44] N. Chen, J. Yue, L. Fang, and S. Xia, "SpectralDiff: A generative framework for hyperspectral image classification with diffusion models," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522416.

[45] J. Zhou et al., "Exploring multi-timestep multi-stage diffusion features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519516.

[46] B. Xi et al., "Deep prototypical networks with hybrid residual attention for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3683–3700, 2020.

[47] K. Gao, B. Liu, X. Yu, J. Qin, P. Zhang, and X. Tan, "Deep relation network for hyperspectral image few-shot classification," *Remote Sens.*, vol. 12, no. 6, p. 923, Mar. 2020.

[48] Z. Xue, Y. Zhou, and P. Du, "S3Net: Spectral–spatial Siamese network for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531219.

[49] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.

[50] C. Li, B. Zhang, D. Hong, J. Yao, and J. Chanussot, "LRR-Net: An interpretable deep unfolding network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513412.

[51] K. Zhang et al., "Spectral–spatial dual graph unfolding network for multispectral and hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508718.

[52] W. Dong, S. Liu, S. Xiao, J. Qu, and Y. Li, "ISPDiff: Interpretable scale-propelled diffusion model for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519614.

[53] D. Hong, C. Li, B. Zhang, N. Yokoya, J. A. Benediktsson, and J. Chanussot, "Multimodal artificial intelligence foundation models: Unleashing the power of remote sensing big data in Earth observation," *Innov. Geosci.*, vol. 2, no. 1, 2024, Art. no. 100055.

[54] D. Hong et al., "Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113856.



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